

Project Title
Legal Document Intelligence Engine
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Legal documents contain large volumes of complex textual information, making efficient searching, comparison, and analysis challenging. The Legal Document Intelligence Engine is an advanced algorithm-based system designed to analyse and retrieve information from a structured corpus of legal documents.

The system applies advanced algorithmic techniques from multiple areas of computer science to provide efficient legal text processing. Exact pattern matching is performed using algorithms such as **Knuth-Morris-Pratt** (KMP) and **Rabin-Karp**, while fuzzy matching techniques based on edit distance can support searches involving spelling variations or incomplete terms. Additional string-processing techniques can be used for multi-pattern search and document similarity analysis.

The project is designed as an integrated legal text analytics system rather than a collection of independent algorithms. It aims to demonstrate the practical application of string algorithms, dynamic programming, graph and network-flow concepts, approximation techniques, and randomised algorithms within a meaningful real-world domain.

By combining efficient algorithms with a structured legal document corpus, the system provides a platform for analysing legal text, comparing algorithm performance, and understanding how advanced algorithmic techniques can be applied to large-scale document processing problems.

Signature of the Guide